Calcite in Equilibrium Example

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1 Chemistry

We consider a reactive transport problem in a one-dimensional column initially containing calcite ($CaCO_3$) in equilibrium with resident water. Clean inflow water, which is in equilibrium with both calcite and the atmosphere (fixed partial pressures of $O_{2(g)}$ and $CO_{2(g)}$), enters from the inflow boundary and mixes with the resident water. This problem tests the ability of the Water Mixing Approach (WMA) to transport reactive solutes in a system governed purely by equilibrium reactions, with no kinetic reactions. The aqueous species considered are H^+, Ca^{2+}, $O_{2(aq)}$, HCO_3^-, and H_2O. The mineral is calcite ($CaCO_3$), which is present as a pure solid phase with constant activity equal one. The gases are $CO_{2(g)}$ and $O_{2(g)}$, which are in equilibrium with the atmosphere at fixed partial pressures.

We consider the following equilibrium reactions:

- Calcite dissolution/precipitation:

$$CaCO_3(s) + H^+ \rightleftharpoons Ca^{2+} + HCO_3^- \qquad (\log_{10} K = 1.8487 \text{ at } 25°\text{C})$$

- Atmospheric O_2:

$$O_{2(g)} \rightleftharpoons O_{2(aq)} \qquad (\log_{10} K = -2.8980 \text{ at } 25°\text{C})$$

- Atmospheric CO_2:

$$CO_{2(g)} + H_2O \rightleftharpoons H^+ + HCO_3^- \qquad (\log_{10} K = -7.8136 \text{ at } 25°\text{C})$$

There are no kinetic reactions in this problem ($\mathbf{S}_K = \mathbf{0}$). Since $CaCO_3(s)$, $O_{2(g)}$, $CO_{2(g)}$, and H_2O all have constant activity, the variable activity species are H^+, Ca^{2+}, $O_{2(aq)}$, and HCO_3^-.

In the domain (column interior), only the mineral calcite is present and there is no gas phase, so only the calcite reaction applies. The equilibrium stoichiometric matrix in the domain, restricted to variable activity species, is

$$\mathbf{S}_{e,v}^{dom} = \begin{array}{c} \begin{matrix} H^+ & Ca^{2+} & O_{2(aq)} & HCO_3^- \end{matrix} \\ \begin{pmatrix} -1 & 1 & 0 & 1 \end{pmatrix} \end{array},$$

where the single row corresponds to calcite dissolution/precipitation: consuming one H^+, producing one Ca^{2+}, leaving $O_{2(aq)}$ unaffected, and producing one HCO_3^-.

At the inflow boundary, the water is in equilibrium with both calcite and the atmosphere, so all three equilibrium reactions apply. The full stoichiometric matrix at the inflow, restricted to variable activity species, is

$$\mathbf{S}_{e,v}^{inf} = \begin{array}{c} \begin{matrix} H^+ & Ca^{2+} & O_{2(aq)} & HCO_3^- \end{matrix} \\ \begin{pmatrix} -1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \end{array},$$

where rows correspond to the calcite dissolution/precipitation, O_2 gas equilibrium, and CO_2 gas equilibrium reactions, respectively.

1.1 Component matrix

Following De Simoni et al. (2005), the component matrix $\mathbf{U}$ is defined such that $\mathbf{U}\mathbf{S}_{e,v}^T = \mathbf{0}$, ensuring that the components $\mathbf{u} = \mathbf{U}\mathbf{c}$ are conservative quantities not affected by equilibrium reactions. Splitting $\mathbf{S}_{e,v} = \left(\mathbf{S}'_{e,v} \mid \mathbf{S}''_{e,v}\right)$ into blocks corresponding to primary and secondary variable activity species, and redefining so that $\mathbf{S}''_{e,v} = -\mathbf{I}$, the component matrix takes the form

$$\mathbf{U} = \left(\mathbf{I}_{N_u} \mid {\mathbf{S}'_{e,v}}^T\right), \tag{1.1}$$

where $N_u = N_v - N_r$ is the number of components, N_v is the number of variable activity species and N_r the number of equilibrium reactions.

Component matrix of the domain. In the domain, we have $N_v = 4$ variable activity species (H^+, Ca^{2+}, $O_{2(aq)}$, HCO_3^-) and $N_r = 1$ equilibrium reaction (calcite dissolution/precipitation), giving $N_u = 4 - 1 = 3$ components. We choose HCO_3^- as the secondary species (it has coefficient $+1$ in the calcite dissolution/precipitation reaction), so the primary species are H^+, Ca^{2+}, and $O_{2(aq)}$.

The secondary block is $\mathbf{S}''^{dom}_{e,v} = (1)$, which we redefine to $-\mathbf{I}$ by multiplying the calcite row by -1. This yields the transformed primary block $\mathbf{S}'^{dom}_{e,v} = (1, -1, 0)$ (coefficients of H^+, Ca^{2+}, $O_{2(aq)}$ after the sign change). The component matrix is therefore $\mathbf{U}^{dom} \in \mathbb{R}^{3\times4}$:

$$\mathbf{U}^{dom} = \begin{pmatrix} H^+ & Ca^{2+} & O_{2(aq)} & HCO_3^- \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

It is straightforward to verify that $\mathbf{U}^{dom}(\mathbf{S}^{dom}_{e,v})^T = \mathbf{0}$. The three conservative components are

$$\begin{aligned} u_1^{dom} &= c_{H^+} + c_{HCO_3^-}, \\ u_2^{dom} &= c_{Ca^{2+}} - c_{HCO_3^-}, \\ u_3^{dom} &= c_{O_{2(aq)}}. \end{aligned}$$

The physical interpretation is as follows: calcite dissolution/precipitation consumes one H^+ and produces one HCO_3^-, so their sum u_1 is conserved; it produces equal moles of Ca^{2+} and HCO_3^-, so their difference u_2 is conserved; and it does not affect $O_{2(aq)}$, so u_3 is trivially conserved.

Component matrix of the boundary (inflow) water. At the inflow, all three equilibrium reactions apply, so $N_r = 3$. With $N_v = 4$ variable activity species, we have $N_u = 4 - 3 = 1$ component. We choose Ca^{2+}, $O_{2(aq)}$, and HCO_3^- as the secondary species (one per reaction), leaving H^+ as the sole primary species.

The secondary block (columns Ca^{2+}, $O_{2(aq)}$, HCO_3^- of $\mathbf{S}^{inf}_{e,v}$, in that order) is

$$\mathbf{S}''^{inf}_{e,v} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which is not $-\mathbf{I}$. We redefine $\mathbf{S}''_{e,v} = -\mathbf{I}$ by premultiplying by $P = -(\mathbf{S}''^{inf}_{e,v})^{-1}$. Since $\det(\mathbf{S}''^{inf}_{e,v}) = -1$, we obtain

$$P = -(\mathbf{S}''^{inf}_{e,v})^{-1} = \begin{pmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

The transformed primary block (column H^+ of $\mathbf{S}^{inf}_{e,v}$) is obtained as $P \cdot (-1, 0, 1)^T = (2, 0, -1)^T$. The component matrix at the inflow is $\mathbf{U}^{inf} \in \mathbb{R}^{1\times4}$. Written out in the natural species order (H^+, Ca^{2+}, $O_{2(aq)}$, HCO_3^-), since the null-space condition $\mathbf{U}^{inf}(\mathbf{S}^{inf}_{e,v})^T = \mathbf{0}$ yields $\mathbf{u} = [a, 2a, 0, -a]$ (up to a scalar), we take $a = 1$ and obtain

$$\mathbf{U}^{inf} = \begin{pmatrix} H^+ & Ca^{2+} & O_{2(aq)} & HCO_3^- \\ 1 & 2 & 0 & -1 \end{pmatrix}.$$

It is straightforward to verify that $\mathbf{U}^{inf}(\mathbf{S}^{inf}_{e,v})^T = \mathbf{0}$: for the calcite row, $1 \cdot (-1) + 2 \cdot 1 + 0 \cdot 0 + (-1) \cdot 1 = 0$; for the O_2 row, $1 \cdot 0 + 2 \cdot 0 + 0 \cdot (-1) + (-1) \cdot 0 = 0$; for the CO_2 row, $1 \cdot 1 + 2 \cdot 0 + 0 \cdot 0 + (-1) \cdot 1 = 0$. The single component is

$$u_1^{inf} = c_{H^+} + 2\,c_{Ca^{2+}} - c_{HCO_3^-},$$

which reflects the charge balance and electroneutrality structure of a solution simultaneously in equilibrium with calcite, atmospheric O_2, and atmospheric CO_2. In that equilibrium state, the concentrations of Ca^{2+}, $O_{2(aq)}$, and HCO_3^- are all fixed by the three equilibrium constraints, leaving H^+ as the only free variable, and hence only one independent component.

Note on the component matrix used in the simulation. Although two component matrices have been derived above ($\mathbf{U}^{dom}$ and $\mathbf{U}^{inf}$), the reactive transport simulation uses the *domain* component matrix $\mathbf{U}^{dom}$, not the boundary one. This is because the components $\mathbf{u} = \mathbf{U}\mathbf{c}$ must be conservative throughout the model domain, where only the calcite dissolution/precipitation equilibrium is active; the additional atmospheric gas equilibria at the inflow are not enforced inside the aquifer. The boundary component matrix $\mathbf{U}^{inf}$ is used only to speciate the inflow water (i.e. to compute the boundary concentrations consistent with calcite and atmospheric equilibrium), and the resulting concentrations $\mathbf{c}_{inf}$ are then projected onto the domain components via $\mathbf{u}_{inf} = \mathbf{U}^{dom}\mathbf{c}_{inf}$ before being imposed as a boundary condition for the transport of components. More generally, the component concentrations $\mathbf{u} = \mathbf{U}^{dom}\mathbf{c}$ used in the reactive transport simulation are *always* computed with the domain component matrix, regardless of whether the underlying species concentrations $\mathbf{c}$ correspond to the domain (initial or interior) or to the boundary (inflow) water.

The initial and boundary concentrations we consider are in Table 1:

	initial (resident)	inflow (boundary)
H^+	10^{-5}	10^{-7}
Ca^{2+}	10^{-3}	10^{-9}
$O_{2(aq)}$	10^{-6}	$2.5295 \cdot 10^{-4\,\dagger}$
HCO_3^-	1	$3.5 \cdot 10^{-4\,\ddagger}$

Table 1: Primary species concentrations of initial and boundary waters (in mol/kg_w). †Constrained by $O_{2(g)}$ equilibrium (partial pressure = 0.2). ‡Constrained by $CO_{2(g)}$ equilibrium (partial pressure = 3.5×10^{-4}).

2 Transport

We solve the following set of equations:

$$\begin{cases} \phi \dfrac{\partial \mathbf{c}^T}{\partial t} = \mathcal{L}^T(\mathbf{c}) + \phi \mathbf{r}^T \\ \mathbf{S}_e \log \mathbf{c} = \log \mathbf{K}_e \\ \mathbf{r}_K = \mathbf{r}_K(\mathbf{c}) \end{cases} \tag{2.1}$$

where $\phi \in (0, 1]$ is porosity; $\mathbf{c} \in \mathbb{R}^{n_s}$ is the concentration array of all species (n_s is the number of species); $\mathcal{L}^T(\mathbf{c}) = -\mathbf{q}^T \nabla \mathbf{c}^T + \nabla \cdot \left(\mathbf{D} \nabla \mathbf{c}^T\right) + f_w \left(\mathbf{c}_r^T - \mathbf{c}^T\right)$ is the transposed advective form of the transport operator, if we solve transport using the advective ADE, where $\mathbf{q} \in \mathbb{R}^d$ is the Darcy flux (d is the Euclidean dimension, in this case $d = 1$), $\mathbf{D} \in \mathbb{R}^{d\times d}$ is the hydrodynamic dispersion tensor, f_w is the (volumetric) water sink-source, with concentration $\mathbf{c}_r$ if $f_w > 0$ (recharge), or, $\mathbf{c}_r = \mathbf{c}$ if $f_w < 0$ (discharge), or $\mathbf{c}_r = 0$ when $f_w < 0$ represents evaporation; $\mathbf{r} = \mathbf{S}_e^T \mathbf{r}_e + \mathbf{S}_K^T \mathbf{r}_K$ is the contribution of chemical reactions to the mass balance of species (per unit time); $\mathbf{S}_e \in \mathbb{R}^{n_e \times n_s}$ and $\mathbf{S}_K \in \mathbb{R}^{n_K \times n_s}$ are the stoichiometric matrices of equilibrium and kinetic reactions, respectively, whose respective reaction rates are given by $\mathbf{r}_e \in \mathbb{R}^{n_e}$ and $\mathbf{r}_K \in \mathbb{R}^{n_K}$ (n_e and n_K are the number of equilibrium and kinetic reactions, respectively) and $\mathbf{K}_e \in \mathbb{R}^{n_e}$ are the equilibrium constants.

Equation (2.1) needs to be solved subject to appropriate initial ($\mathbf{c}(\mathbf{x}, t) = \mathbf{c}(\mathbf{x}, 0)$ throughout the model domain) and boundary conditions, which we choose to be prescribed mass flux, that is,

$$\begin{cases} \mathbf{n}^T \left(\mathbf{q}\mathbf{c} - \mathbf{D}\nabla \mathbf{c}\right) = q_{inf}\mathbf{c}_{inf} & \text{at the inflow} \\ \mathbf{n}^T \left(-\mathbf{D}\nabla \mathbf{c}\right) = 0 & \text{at the outflow} \end{cases}$$

expressing that the flux of solute mass at the inflow is $q_{inf}\mathbf{c}_{inf}$, and that there is no dispersive flux at the outflow.

To solve numerically, we choose a uniform mesh made of 10 cells of length $\Delta x = 0.1$ m (domain length $L = 1$ m). The time step is $\Delta t = 10^{-2}$ days. We use the fully implicit Euler scheme for both transport and reactions. We consider uniform transport properties: $\phi = 0.5$, $q = q_{inf} = 0.2m/d$, $\alpha_L = 1m$. The dispersion coefficient is $D = \alpha_L q = 0.2m^2/d$. Note that while the chemistry described in Section 1 (species, reactions, stoichiometry, and equilibrium constants) is intrinsic to this calcite equilibrium problem, the transport parameters and numerical settings — mesh size, time step, porosity, Darcy flux, dispersivity, boundary condition type, and integration scheme — can be freely modified by the user to suit different physical scenarios, subject to the following stability and accuracy constraints:

- **Grid Péclet number:** $Pe = \frac{q\Delta x}{D} = \frac{\Delta x}{\alpha_L} \leq 2$ to avoid spurious oscillations in centred schemes. In this example, $Pe = 0.1$.
- **Courant number:** $Co = \frac{q\Delta t}{\phi \Delta x} \leq 1$ for time integration. In this example, $Co = 0.2 \cdot 10^{-2}/(0.5 \cdot 0.1) = 0.04$.
- **Von Neumann stability:** For the explicit Euler scheme, the time step must satisfy $\Delta t \leq \frac{\phi(\Delta x)^2}{2D}$ for dispersion stability. The fully implicit Euler scheme is unconditionally stable, so this constraint does not apply.

- **Combined constraint (explicit Euler):** When using explicit time integration, both the Courant and dispersion constraints must be satisfied simultaneously: $\Delta t \leq \min\left(\frac{\phi \Delta x}{q}, \frac{\phi (\Delta x)^2}{2D}\right)$.

The fully implicit Euler scheme used in this example is unconditionally stable, allowing larger time steps at the cost of solving a nonlinear system at each step. However, accuracy still requires Δt small enough to resolve the reaction dynamics.

References

De Simoni, M., Carrera, J., Sanchez-Vila, X., and Guadagnini, A. (2005). A procedure for the solution of multicomponent reactive transport problems. *Water resources research*, 41(11).